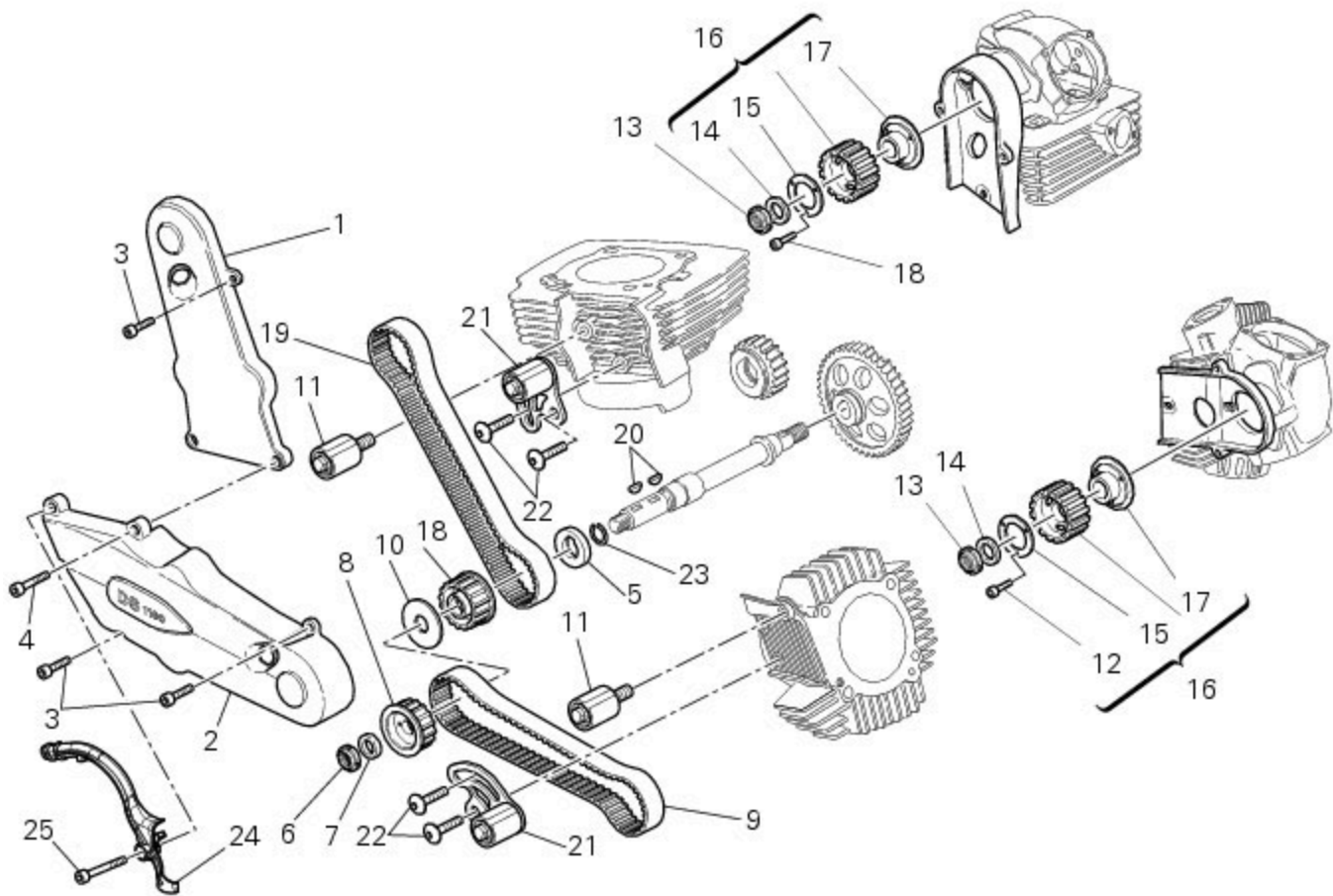


4.2 -Cylinder head assemblies: timing belt covers/timing system



- 1 Vertical cylinder timing belt cover
- 2 Horizontal cylinder timing belt cover
- 3 Screw
- 4 Screw
- 5 Spacer
- 6 Elastic stop ring
- 7 Washer
- 8 Outer timing belt driveshaft pulley
- 9 Horizontal cylinder timing belt
- 10 Intermediate spacer
- 11 Idler pulley assembly
- 12 Screw
- 13 Elastic stop ring
- 14 Washer
- 15 Special washer
- 16 Camshaft pulley
- 17 Flange
- 18 Inner timing belt driveshaft pulley
- 19 Vertical cylinder timing belt
- 20 Key
- 21 Tensioner pulley assembly
- 22 Screw
- 23 Circlip
- 24 Guide
- 25 Screw

1100 ABS	TIMING SYSTEM
1100 ABS	HORIZONTAL CYLINDER HEAD
1100 S ABS	TIMING SYSTEM
1100 S ABS	HORIZONTAL CYLINDER HEAD

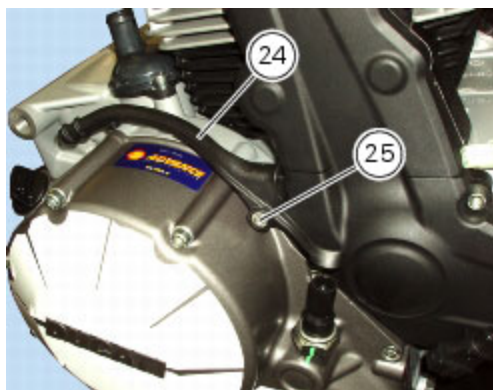
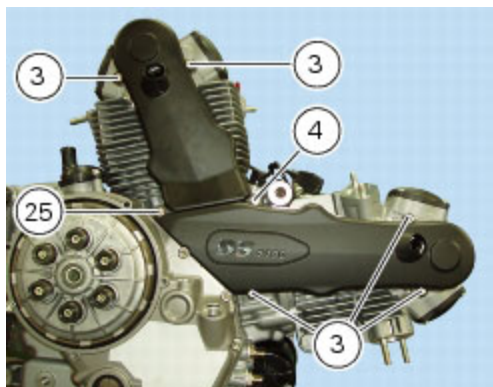
Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the timing belt covers

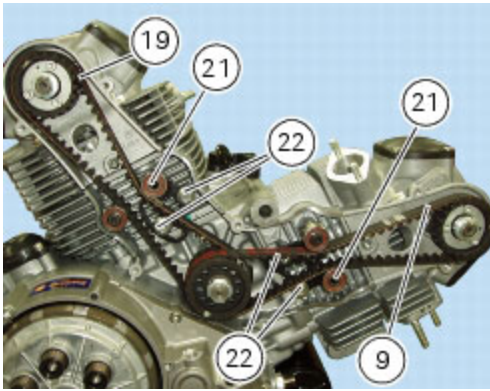
Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Disconnect the right-hand side spark plug cables	M 3, Spark plug
Remove the horizontal cylinder exhaust pipe	L 8, Removal of the exhaust system

Unscrew the screw (4), screws (3), (25) and remove the horizontal timing belt cover (2).
Undo the screw (4) securing the vertical timing belt external cover (1) and remove it.
Loosen screw (25) to remove cable grommet (24).



Removal of the timing system assembly

Loosen the positioning bolt (22) of the tensioner pulley (21) of the horizontal cylinder timing belt.
To move the tensioner pulley (21) to the rest position.
Remove the horizontal cylinder timing belt (9).
Repeat the above operations to remove the vertical cylinder timing belt (19).



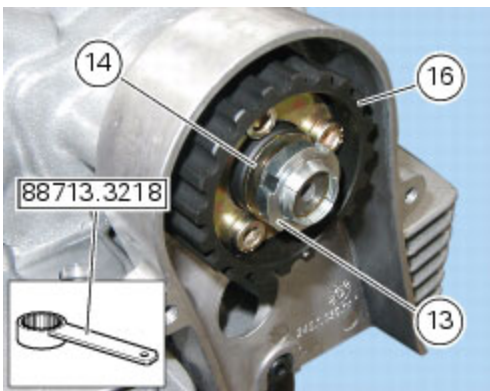
To remove the pulleys from the cylinder heads, install tool no. **88713.3218** to the pulley to lock out its rotation, then loosen ring nut (13).

Remove the nut (13), washer (14) and pulley (16).



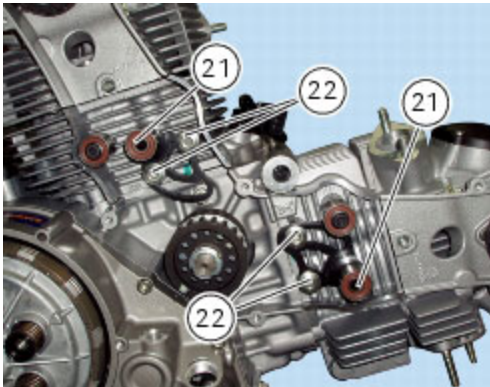
Important

On reassembly, always fit new nuts.



Remove the bolts (22) securing the timing belt tensioner pulleys (21).

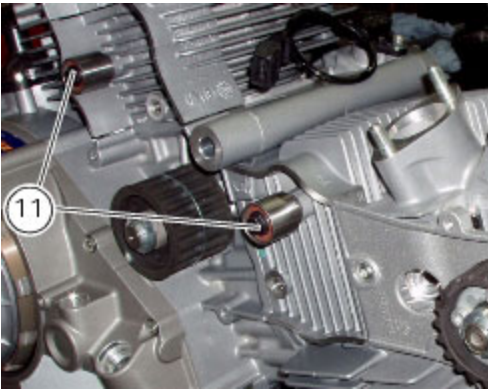
Remove the two timing belt tensioner pulleys (21).



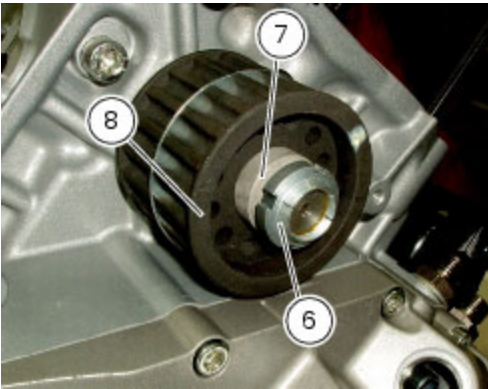
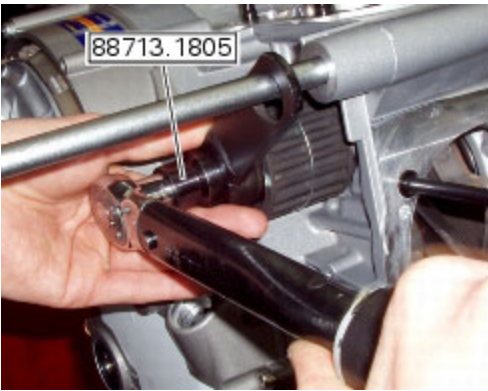
Note

The tensioner pulleys can be also be removed from the engine without removing the timing belts.

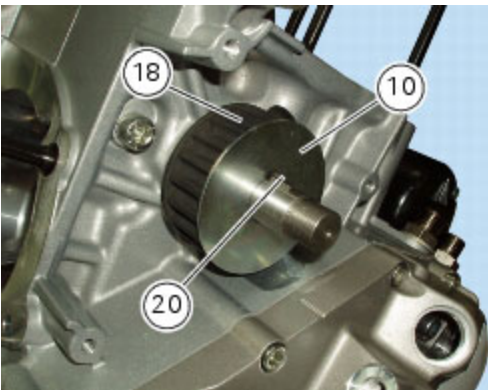
Unscrew and remove the idler pulleys (11).



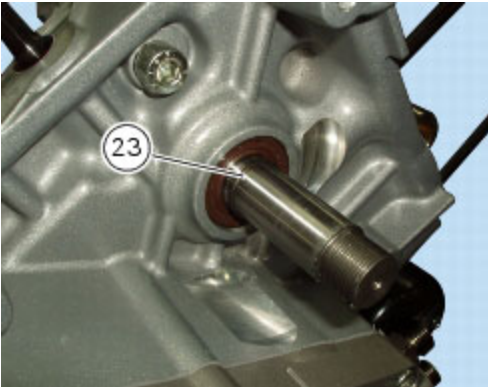
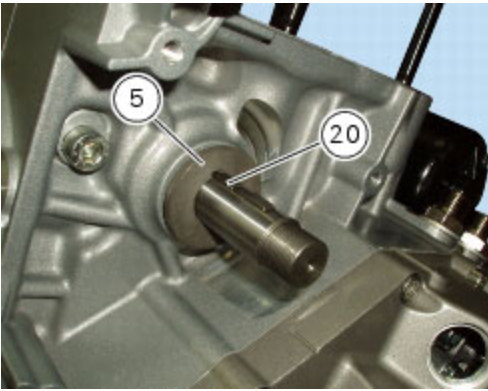
Use the wrench of tool no. **88713.1805** to lock rotation of the timing belt driveshaft pulley on the crankcase and, using the socket combined with the tool, loosen the ring nut (6). Remove the nut (6), the washer (7) and the outer pulley (8).



Remove the first Woodruff key (20) located on the timing belt driveshaft. Remove the intermediate spacer (10) and the inner pulley (18).



Remove the second Woodruff key (20) located on the timing belt driveshaft. Remove the spacer (5) and the circlip (**23**).

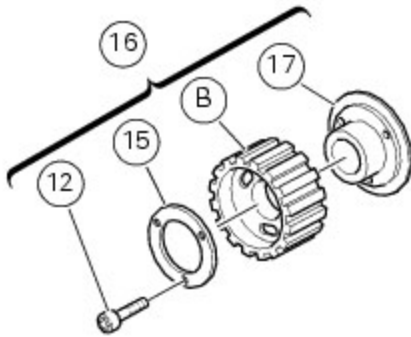


Disassembly of the pulleys

Loosen and remove the three bolts (12).

Slide off the washer (15).

Withdraw the pulley (B) from the flange (17).

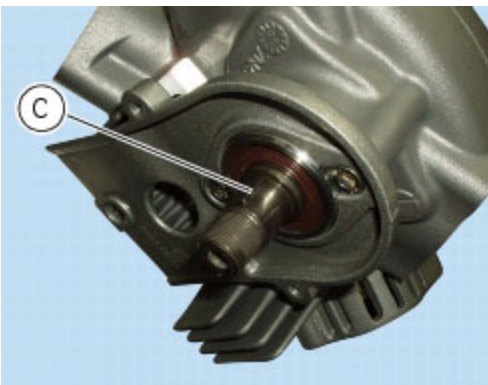


Refitting the timing system assembly

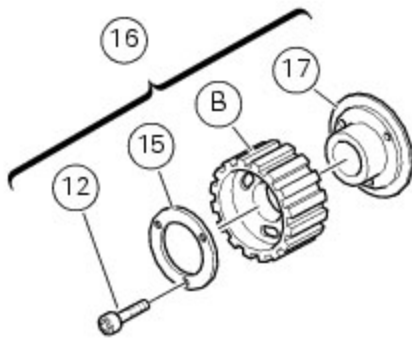
Refitting the camshaft timing pulleys

Check that the keyway on the end of camshaft is in good condition and without burrs.

Fit one key (C) into the camshaft keyway.

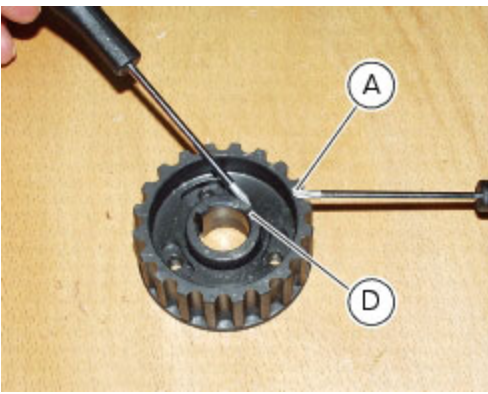


To assemble these parts, install the pulley (B) and washer (15) in the flange (17), as shown in the figure.



Important

During reassembly, take care that the timing mark "D" on the spacer flange is aligned with the mark "A" on the pulley and washer.



Screw the three bolts (12) into the flange.
Fit the assembled pulley (16) onto the camshaft, so that is firmly seated.

Warning

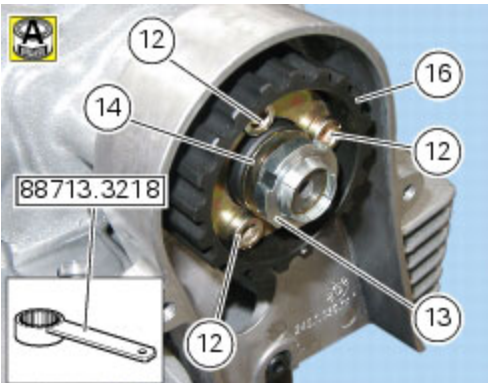
Make sure that the tongue is not dislodged.

Insert tool no. **88713.3218** in the belt pulleys to prevent their rotation.
Apply a smear of grease to the thread and underside of the nut (13), and fit the washer (14) and the nut.

Important

Always fit new nuts on reassembly.

Tighten the ring nuts (13) to the specified tightening torque (Sect. C 3, [Engine torque settings](#)).

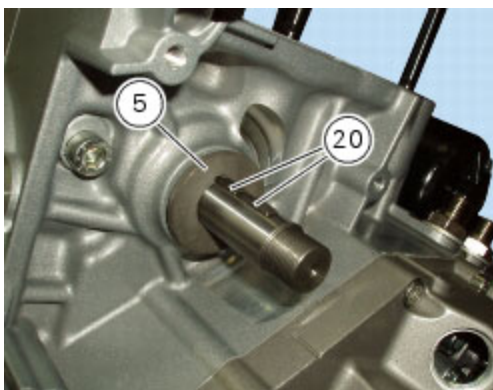


Refitting the timing belt driveshaft pulleys

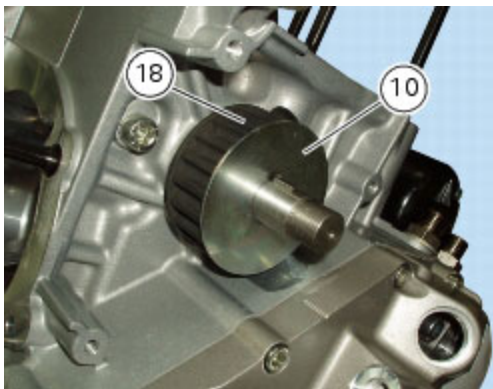
Install the circlip (23) on the pulley side of the timing belt driveshaft. Drive it into its seat using the protective cap **88700.5749**.



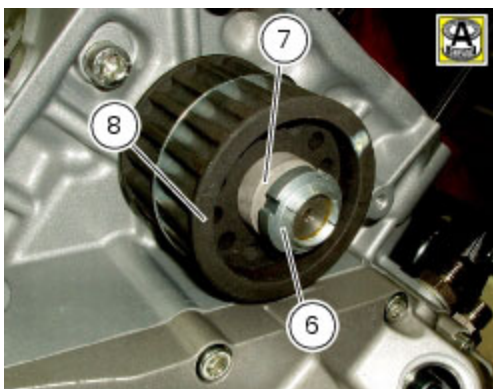
Fit the two Woodruff keys (20) and the spacer (5) with the slot facing the crankcase.



Locate the inner pulley (18), with the protruding hub facing outwards, and the guide washer (10) on the timing belt driveshaft.



Then fit the outer pulley (8) with the protruding hub facing inwards, outer spacer (7) and lock nut (6), suitably lubricated.



To prevent the nuts working loose and consequent serious engine damage, always use new self-locking nuts on all the timing belt pulleys on reassembly.

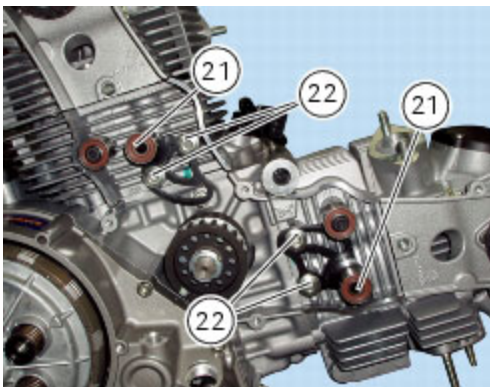
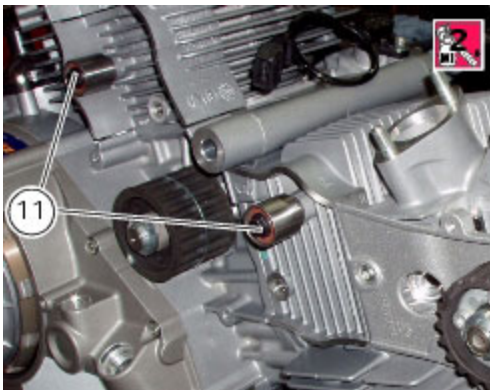
Use the wrench of tool **88713.1805** to restrain the pulleys against rotation and, using the bush of the tool in a torque wrench, tighten the self-locking nut to the specified torque (Sect. C 3, [Engine torque settings](#)).



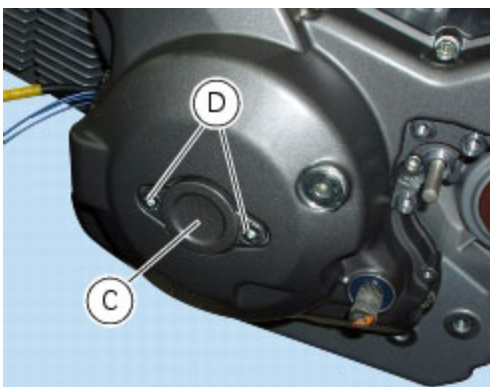
Refitting the timing belt tensioner pulleys

If removed, apply the recommended threadlocker to the idler pulley (11) and tighten to the specified torque (Sect. C 3, [Engine torque settings](#)).

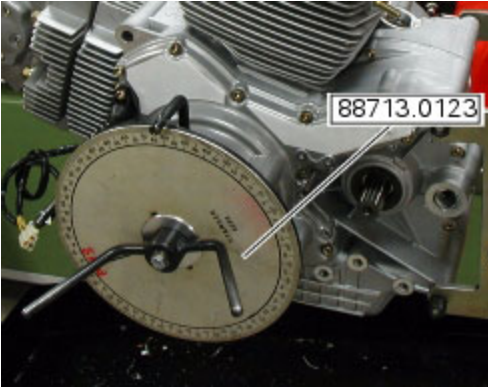
Locate the tensioner pulleys (21) on the vertical and horizontal cylinders. Screw in bolt (22) without fully tightening it.



Remove the inspection cover (C) from generator cover by unscrewing the two screws (D).



Install service tool no. **88713.0123**.

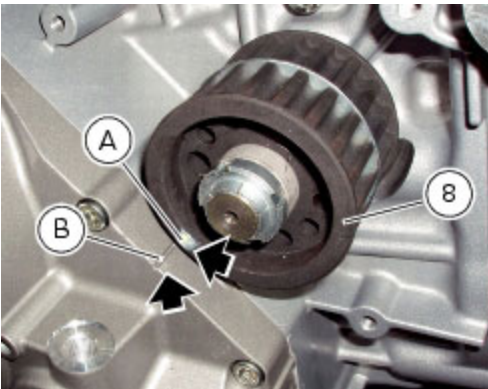


Position the crankshaft so that the piston of the horizontal cylinder is at TDC of its power stroke, turn it using the tool no. **88713.0123**. This is achieved by aligning the timing mark (A) on the timing belt driveshaft pulley (8) with the reference notch (D) on the crankcase cover.



Important

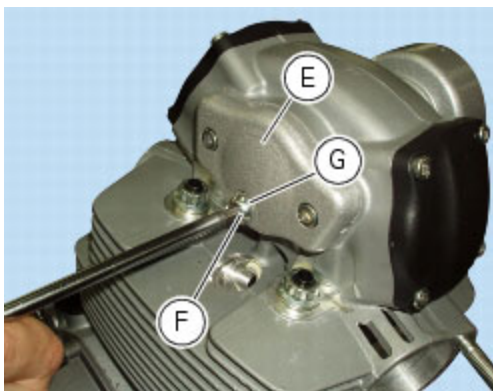
Turn the timing belt driveshaft pulley with the spark plugs removed from the engine.



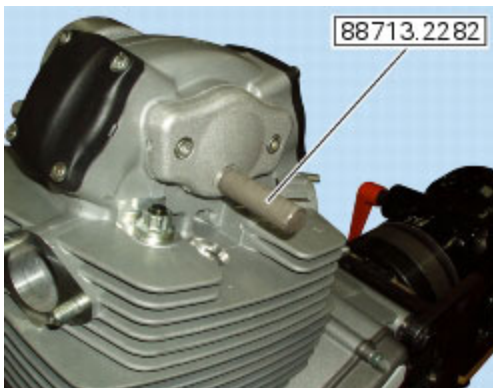
Remove the tool no. **88713.0123** and install in the generator cover seat the tool no. **88713.2011** to prevent crankshaft rotation and lock it with the corresponding pin.



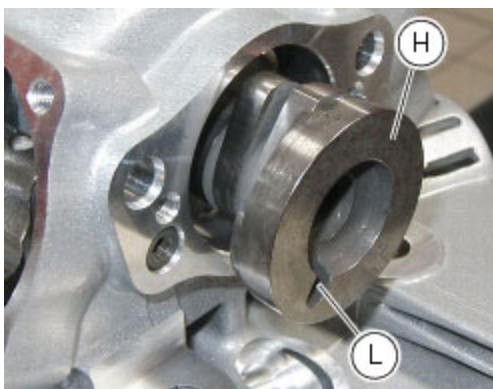
Remove the bolt (F) from the inner timing belt cover (E) and recover the washer (G).



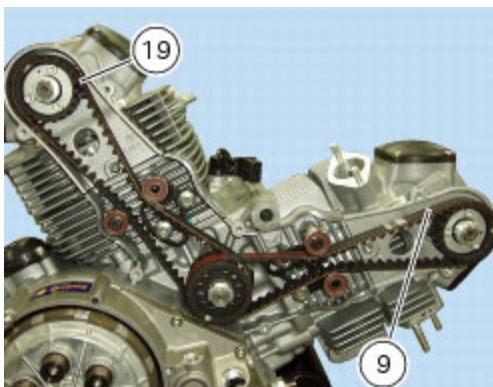
Restrain the camshaft (H) of both cylinder heads by screwing tool no. **88713.2282** into the side covers (E).



Turn the pulleys by hand as shown in the figure until the end of the tool engages the slot (L) on the camshaft.



Fit the horizontal (9) and vertical (19) timing belts using your hands only and without altering the alignment of the pulleys.

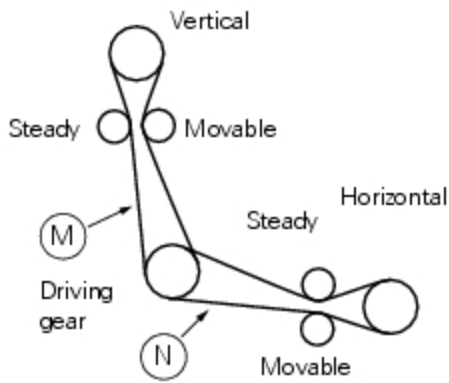


Check the timing belt tension and adjust if necessary as described in the paragraph "[Measuring timing belt tension](#)" (Sect. D 5).

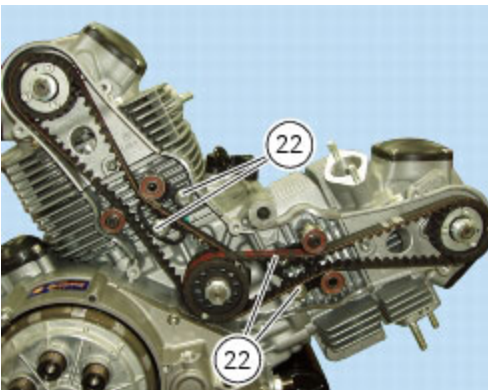
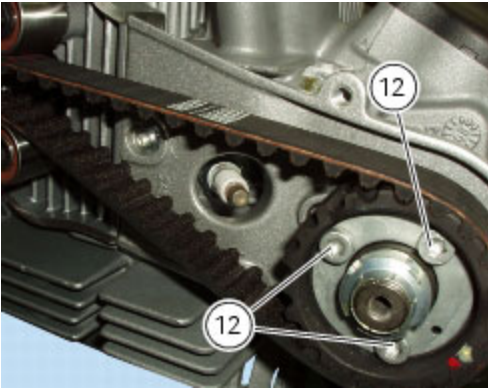


Warning

Check the tension values on sections of belts (M) and (N) shown in the figure.



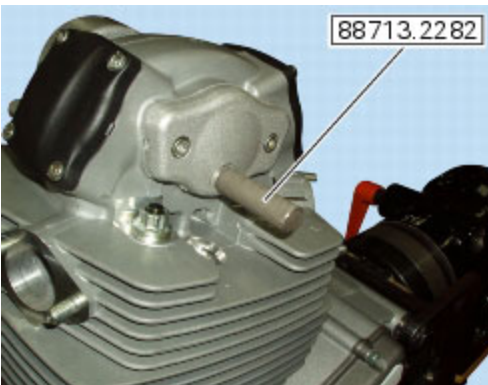
When the correct belt tensioning value is reached, ensure that screws (22) and (12) are tightened to the specified torque (Sect. C 3, [Engine torque settings](#)).

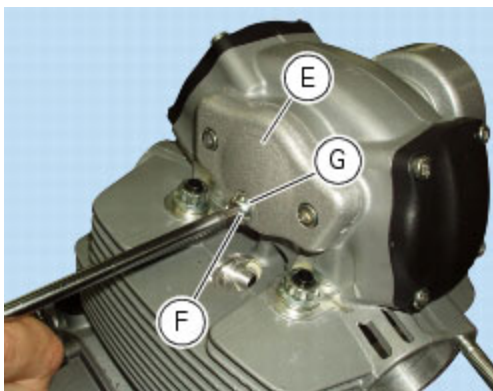


Warning

Whenever you renew any components of the timing system, it is good practice to also check the timing (Sect. N 4.1, [Cylinder head assemblies: checks and adjustments](#)).

Remove the service tools no. **88713.2282** from the side covers (E) and refit the screws (F) with washer (G), tightening them to the specified torque (Sect. C 3, [Engine torque settings](#)).





Remove the optical reader and the tester.

Remove the crankshaft locking tool **88713.2011** unscrewing first the pin and then the screw from the tool.



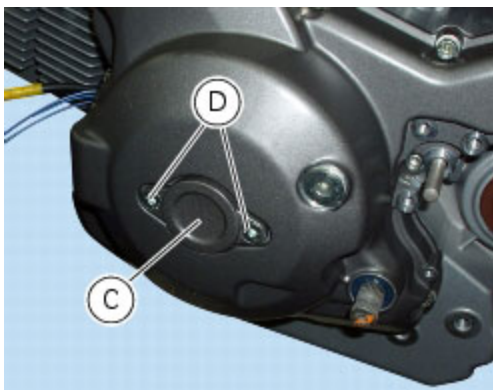
Refit the inspection cover (C) in the generator cover, inserting the two screws (D).

Tighten the screws to the specified torque (Sect. C 3, [Engine torque settings](#)).



Note

Before refitting the cover (C), check the condition of the O-ring and renew it if necessary.



Refitting the timing belt covers

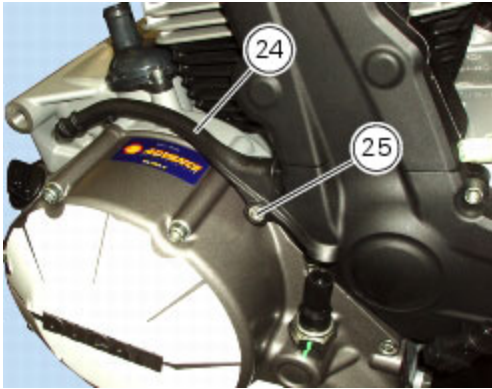
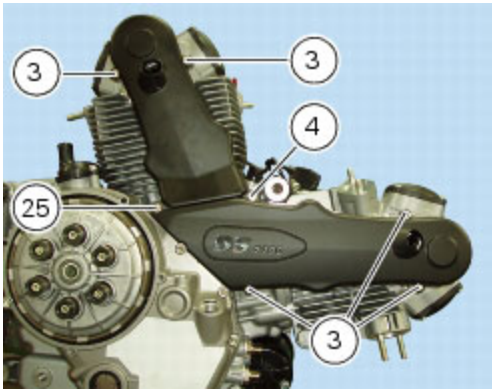
Position the vertical timing belt cover (1) on the engine.

Fit the screws (3) to the cover and tighten them to the specified torque (Sect. C 3, [Engine torque settings](#)).

Locate the horizontal timing belt cover (2) on the engine.

Fit the cable guide (24) on the cover (2) starting screw (25) in its thread.

Tighten the screws (3), (4) and (25) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Operation	Section reference
Refit the horizontal cylinder exhaust pipe	L 8, Refitting the exhaust system
Connect the right-hand side spark plug cables	M 3, Spark plug
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat